

LOWER TRUNK

Many of the muscles involved in movements of the lower trunk run between the pelvic bones and either the spinal column or rib cage. The abdominal muscles (external oblique, internal oblique, and rectus abdominis) and quadratus lumborum flex the trunk by pulling the rib cage toward the pelvis. On the other hand, the other trunk flexors, iliacus and psoas major, work by either pulling the thigh bone (femur) toward the pelvis (iliacus) or pulling the spinal column toward the femur (psoas major). The prime trunk extensors (iliocostalis lumborum, longissimus thoracis, and spinalis thoracis) are collectively called the erector spinae. The iliocostalis lumborum runs between the posterior pelvis and posterior spinal column, while the longissimus thoracis and spinalis thoracis run along the posterior spinal column and help the individual vertebrae in the spinal column work together as a single unit. The interspinales, intertransversarii, multifidus, and rotatores run between individual vertebrae and cause large movements by making small changes between individual pairs or groups of vertebrae. Figures showing these muscles as well as a chart showing specific movements for each muscle are located at the end of the chapter (pages 66-67).

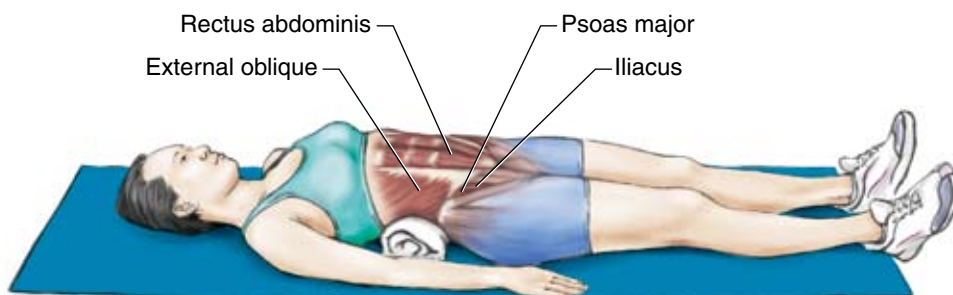
Many people have stiff back muscles and have discovered that stretching helps to relieve some of the pain accompanying stiff back muscles. The back muscles (or trunk extensors) are not the only lower-trunk muscles to influence back pain. Often people can find relief from back pain by leaning backward (trunk hyperextension); this action actually stretches the abdominal muscles (trunk flexors). This shows that flexible trunk flexors are also important. Moreover, numerous sporting activities (such as golf, tennis, and throwing sports) require twisting of the trunk. Twisting the trunk involves the trunk extensors, flexors, and lateral flexors. Improved range of motion of all lower-trunk muscles can increase the range of motion in trunk rotation and improve the performance in activities that involve these actions.

Hyperextension (arching) and hyperflexion of the lower back are potentially dangerous, especially if you have weak abdominal, thigh, and buttocks muscles. Furthermore, backward roll movements are potentially dangerous to the cervical spine (neck). Injuries from the former and latter stretches may involve excessively squeezing the spinal discs, jamming together the spinal joints, and pinching the spinal nerves emerging from the lumbar vertebrae. Hence, if you perform these stretches, you should do these more gradually than most of the other stretches. Also, to keep pressure off of the neck when doing back rolls, keep the shoulder blades in contact with the floor.

Remember that overstretching (very hard stretching) causes more harm than good. Sometimes the muscles become stiff from overstretching. Overstretching can reduce muscle tone; when tone is reduced, the body compensates by making the loose muscle excessively tight. So, for each progression, start with the position that is the least stiff and progress to the next position only when, after several days of stretching, you notice a consistent lack of stiffness during the exercise. This means that you should stretch both the agonist and antagonist muscles. Also, remember that although there may be greater stiffness in one direction (right versus left), you should stretch both sides so that you maintain proper muscle balance.

Some of the instructions and illustrations in this chapter are given for the left side or right of the body. Similar but opposite procedures would be used for the opposite side of the body.

Lower-Trunk Flexor Stretch (Back-Lying Position)



Technique

Lie on the floor on your back.

Place a rolled-up towel (1 to 2 inches, or 2.5 to 5 centimeters, in diameter) between the small of your back and the floor.

Muscles Stretched

Most-stretched muscles: Rectus abdominis, external oblique, internal oblique.

Lesser-stretched muscles: Quadratus lumborum, psoas major, iliacus.

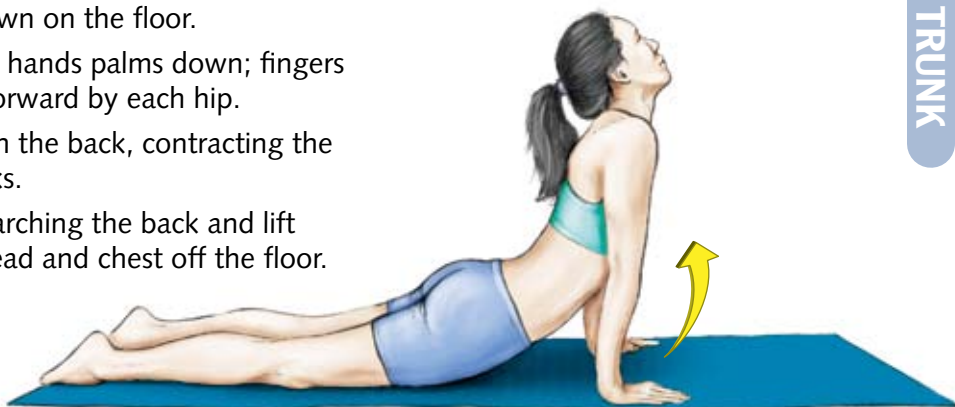
Commentary

Of all of the stretches in this book that stretch the lower trunk flexors, this stretch is the best for people who have a swayed back or have weak abdominal muscles, since arching the lower back is potentially dangerous for these people. Because the small of the back is supported in this exercise, undesired pressures on the spinal column are reduced. Nevertheless, the width of the back support is important. The larger the diameter of the object, the greater the undesired pressure. Make sure that the upper back, shoulder blades, and buttocks are resting comfortably on the floor. Also, squeezing the buttocks will reduce stress on the lower back.

Lower-Trunk Flexor Stretch (Front-Lying Position)

Technique

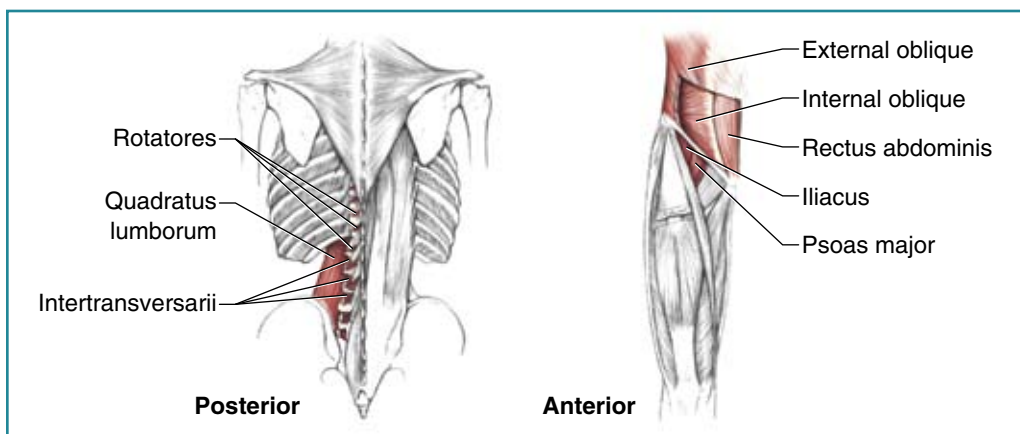
- Lie facedown on the floor.
- Place both hands palms down; fingers point forward by each hip.
- Slowly arch the back, contracting the buttocks.
- Continue arching the back and lift your head and chest off the floor.



Muscles Stretched

Most-stretched muscles: Rectus abdominis, external oblique, internal oblique.

Lesser-stretched muscles: Quadratus lumborum, psoas major, iliacus, rotatores, intertransversarii.



Commentary

Remember that arching the lower back is potentially dangerous, especially if you have weak abdominal muscles. Injuries from arching the lower back include excessive squeezing of the spinal discs, jammed spinal joints, and pinched spinal nerves emerging from the lumbar vertebrae. Therefore, this stretch is recommended only for those who are very stiff. When doing this stretch, do minimal arching and make sure that you squeeze the buttocks during the arching. Squeezing the buttocks reduces stress on the lower back.

Seated Lower-Trunk Extensor Stretch

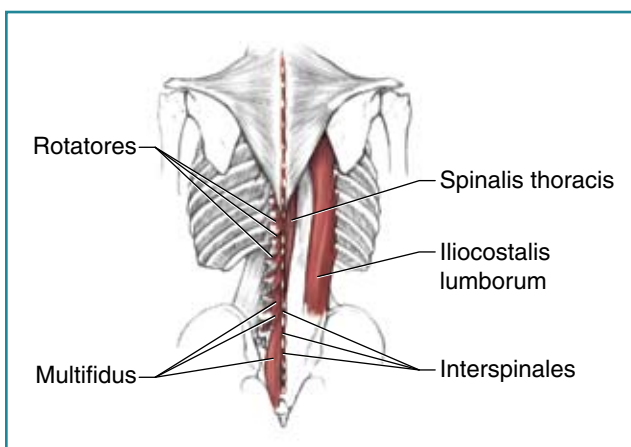


Technique

Sit upright in a chair with legs separated.

Slowly round the upper back and begin to lean forward.

Continue to bend at the waist and lower the head and abdomen between the legs and below the thighs.



Muscles Stretched

Most-stretched muscles: Iliocostalis lumborum, multifidus.

Lesser-stretched muscles: Interspinales, rotatores, spinalis thoracis.

Commentary

Remember that hyperflexion can injure the spinal cord. When doing this exercise, go slowly and do not let the back become straight. Also, the effect of the stretch is minimized if the buttocks rise up off of the chair.

VARIATION

Seated Lower-Trunk Extensor Lateral Flexor Stretch

Angling the head toward one of the knees will increase the stretch on the lower-trunk extensors and partially stretch some of the lateral flexors.

Technique

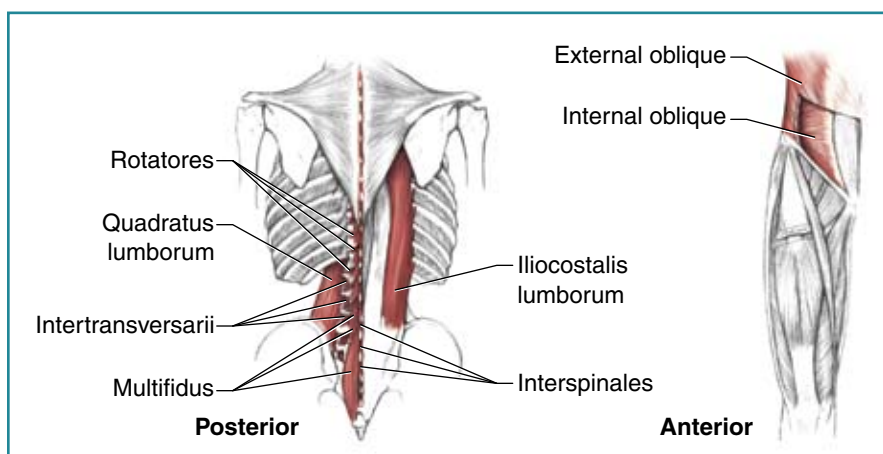
- Sit upright in a chair with legs separated.
- Slowly extend the upper back and begin to lean forward.
- Continue to bend at the waist and lower your head and abdomen toward the right knee.
- Slowly lower the head below the right knee.



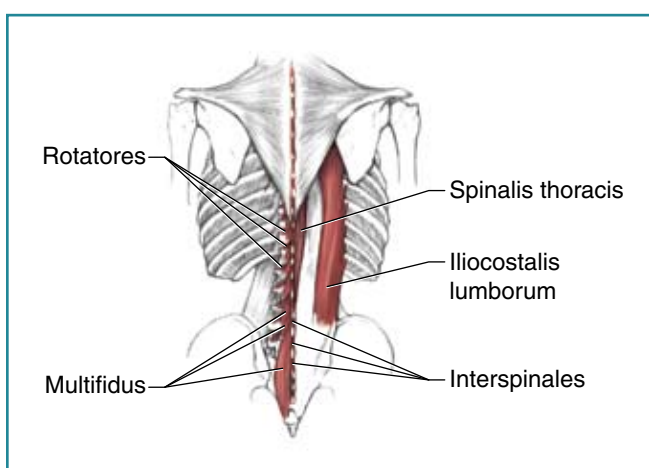
Muscles Stretched

Most-stretched muscles: Left iliocostalis lumborum, left multifidus, left rotatores, left external oblique, left internal oblique.

Lesser-stretched muscles: Left interspinales, left intertransversarii, left quadratus lumborum.



Reclining Lower-Trunk Extensor Stretch



Technique

Lie on the back with the legs extended.

Flex the knees and hips, bringing the knees up over the chest.

Cross the feet at the ankles and separate the knees so that they are at least shoulder-width apart.

Grasp the thighs at the inside of the knees and pull the legs down to the chest.

Muscles Stretched

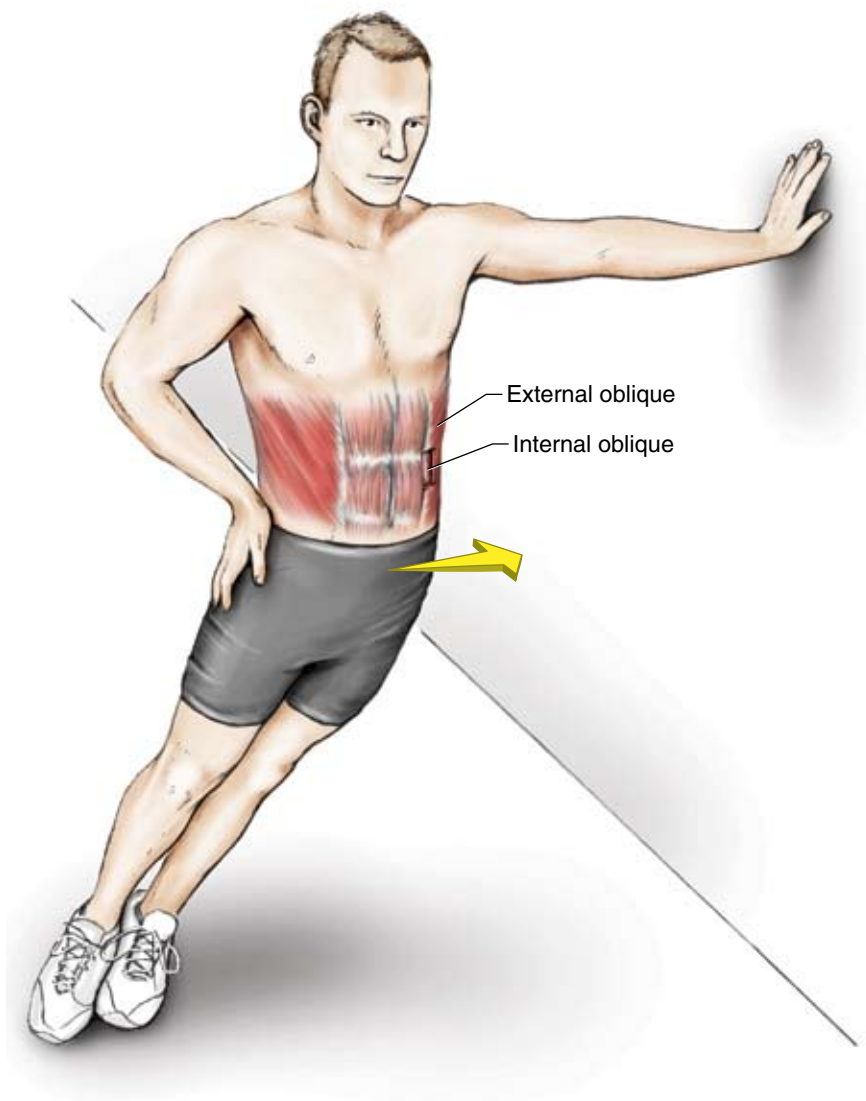
Most-stretched muscles: Iliocostalis lumborum, multifidus.

Lesser-stretched muscles: Interspinales, rotatores, spinalis thoracis.

Commentary

Remember that hyperflexion can injure the spinal cord. When doing this exercise, go slowly and do not let the back become straight. To prevent a straight back, allow the spinal column to curl, and raise the buttocks off the floor. Also, do not try to bring the knees too far below the chest (*do not* try to touch the knees to the floor).

Standing Lower-Trunk Lateral Flexor Stretch



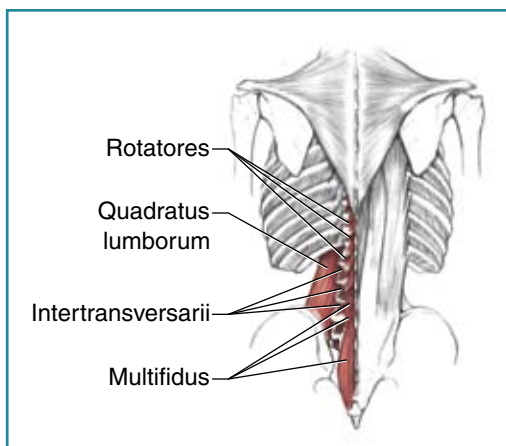
Technique

Stand upright with the feet together and the left side of the body facing a wall about an arm's length away.

Place the palm of the left hand on the wall at shoulder height, and place the heel of the right hand at the hip joint.

While keeping the legs straight, contract the buttocks and slightly rotate the hips in toward the wall.

Use the right hand to push the right hip toward the wall.



Muscles Stretched

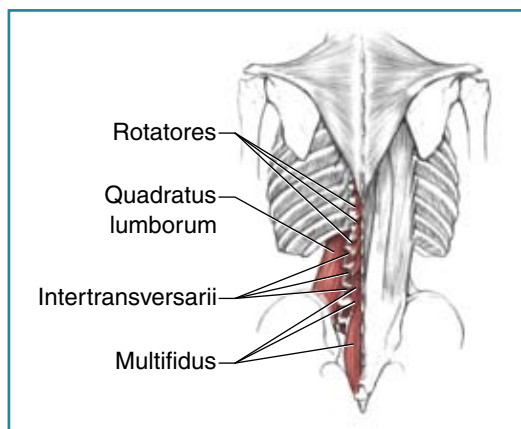
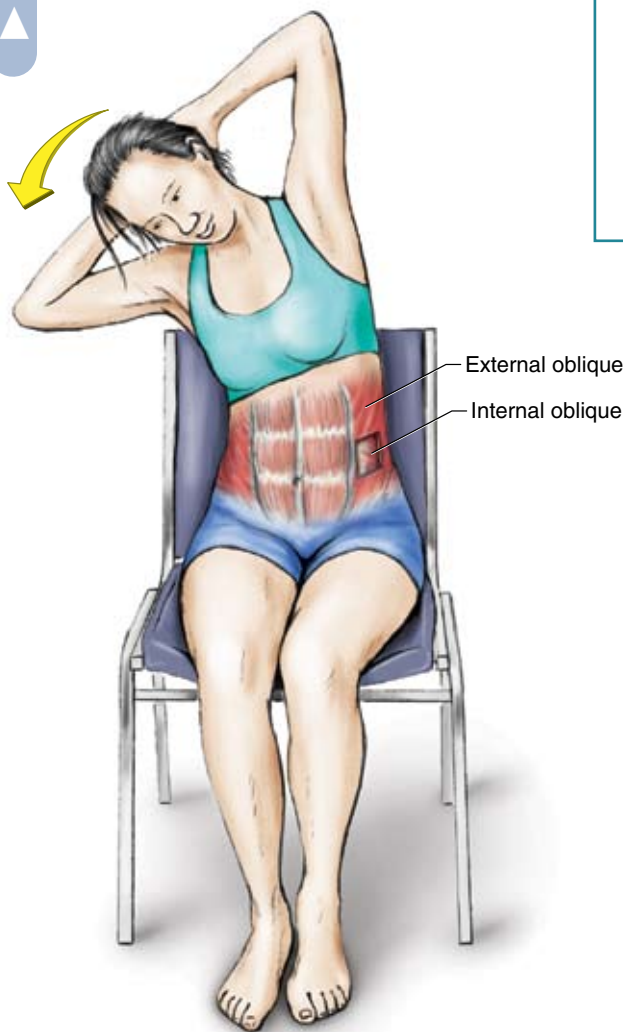
Most-stretched muscles: Left external oblique, left internal oblique, left rotatores.

Lesser-stretched muscles: Left intertransversarii, left multifidus, left quadratus lumborum.

Commentary

It is very easy to lose balance while doing this exercise, so stand on a nonskid surface. Keep the left arm straight, but do not lock the elbow. You can increase the amount of stretch either by moving the feet farther from the wall, by resting the left forearm on the wall instead of the hand, or both.

Seated Lower-Trunk Lateral Flexor Stretch



Technique

Sit upright in a chair.

Interlock the hands behind the head, with the elbows in a straight line across the shoulders.

While keeping both elbows back and in a straight line, laterally flex the waist, and move the right elbow toward the right hip.

Muscles Stretched

Most-stretched muscles: Left external oblique, left internal oblique, left rotatores.

Lesser-stretched muscles: Left intertransversarii, left multifidus, left quadratus lumborum.

Commentary

Flexing or extending at the waist will reduce this stretch's effectiveness. Also, keep the buttocks and thighs in complete contact with the chair. The closer the elbow gets to the floor, the harder it will be to remain seated in the chair. Wrapping the lower legs and feet around the chair legs will help in keeping the buttocks and thighs in contact with the seat.

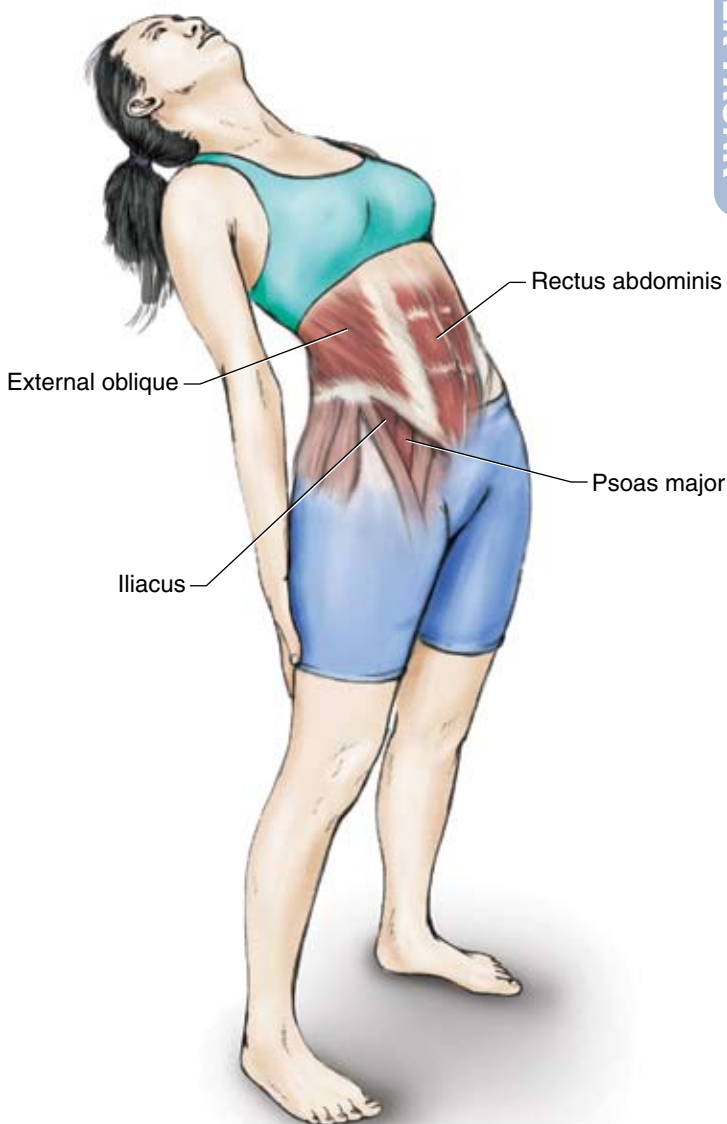
Standing Lower-Trunk Flexor Stretch (Arched Back)

Technique

Stand upright with legs 2 to 3 feet apart (61 to 91 cm) with hands placed on the hips.

Slowly arch the back, contracting the buttocks and pushing the hips forward.

Continue arching the back, drop the head backward, and slide the hands past the buttocks and down the legs.



Muscles Stretched

Most-stretched

muscles: Rectus abdominis, external oblique, internal oblique.

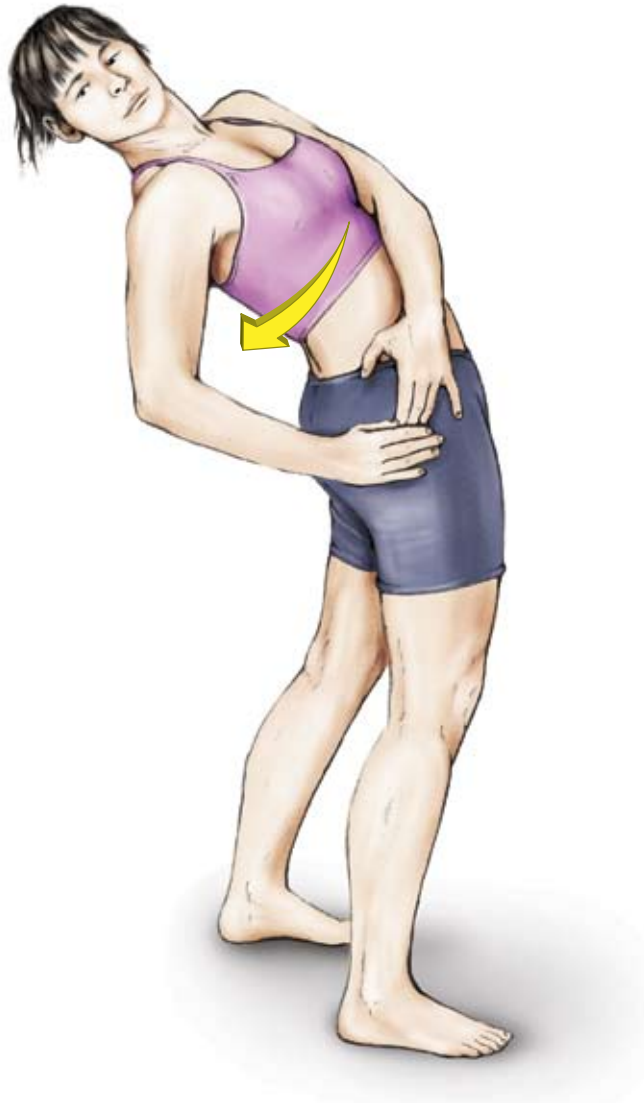
Lesser-stretched

muscles: Quadratus lumborum, psoas major, iliacus.

Commentary

This exercise is potentially dangerous, especially for those who have a swayed back or weak abdominal muscles. This exercise can worsen a swayed back and cause excessive squeezing of the spinal discs, jammed spinal joints, and pinched spinal nerves emerging from the lumbar vertebrae. This stretch is recommended only for those who are very stiff and do not have a swayed back. Also, you should use this exercise only when the other lower-back flexor stretches do not provide any improvement. When doing this stretch, do minimal arching and make sure that you squeeze the buttocks during the arching. Squeezing the buttocks reduces the stress on the lower back.

Lower-Trunk Lateral Flexor Stretch (Arched Back)



Technique

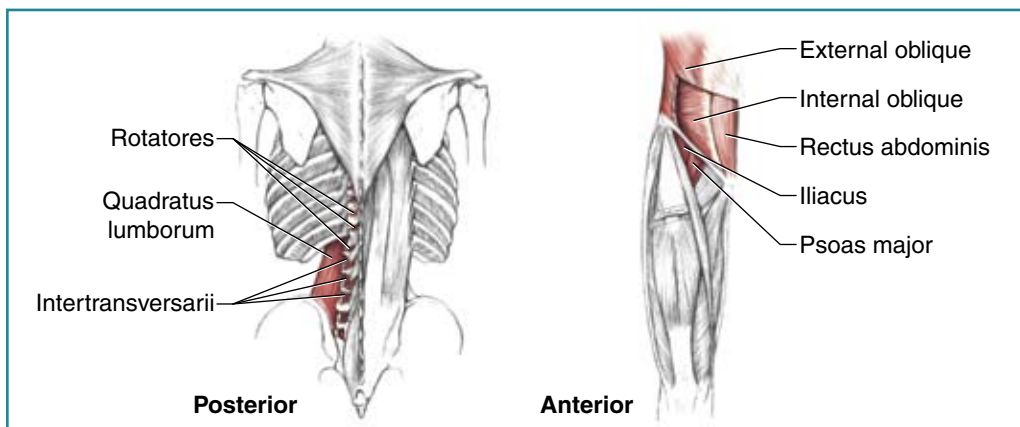
Stand upright with legs 2 to 3 feet apart (61 to 91 cm) with the right foot about 1 foot (30 cm) ahead of the left foot.

Place both hands near the right hip.

Slowly arch the back, contracting the buttocks and pushing the hips forward.

Continue arching the back, rotate the trunk clockwise, and drop the head back toward the right side.

Slide the hands past the right buttock and down the right leg.



Muscles Stretched

Most-stretched muscles: Rectus abdominis, left external oblique, left internal oblique.

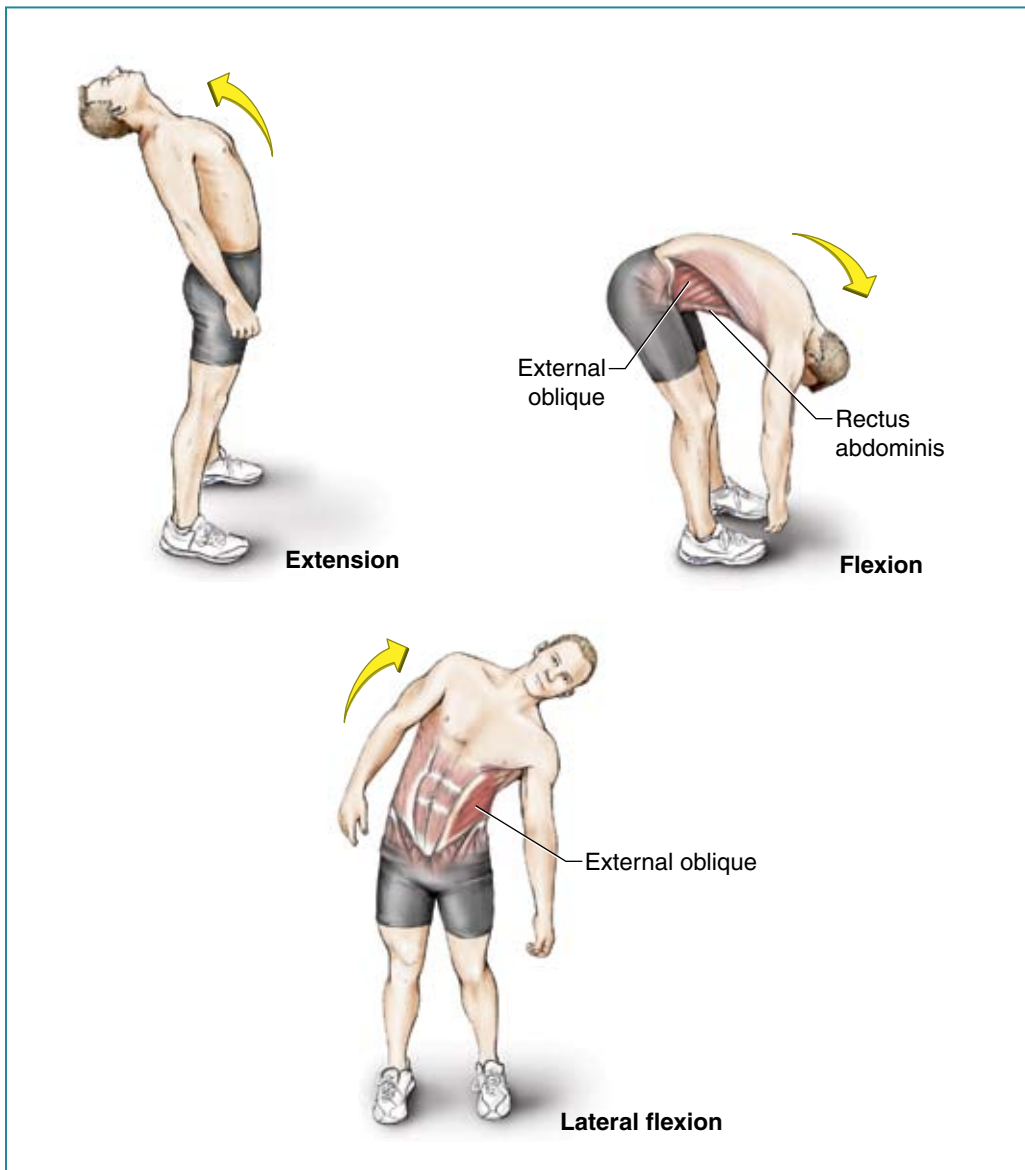
Lesser-stretched muscles: Left quadratus lumborum, left psoas major, left iliacus, left rotatores, left intertransversarii.

Commentary

This exercise is potentially dangerous, especially for people with a swayed back or weak abdominal muscles. This exercise can worsen a swayed back and cause excessive squeezing of the spinal discs, jammed spinal joints, and pinched spinal nerves emerging from the lumbar vertebrae. This stretch is recommended only for those who are very stiff and do not have a swayed back. Also, you should use this exercise only when the other lower-back flexor stretches do not provide any improvement. When doing this stretch, do minimal arching and make sure that you squeeze the buttock during the arching. Squeezing the buttock reduces the stress on the lower back. Finally, it is very easy to lose balance while doing this exercise, so take extra care.

Lower-Trunk Muscle Movements

The stretches in this chapter are excellent overall stretches; however, not all of these stretches may be completely suited to each person's needs. The muscles involved in the various lower-trunk movements appear in the following table. To stretch specific muscles, the stretch must involve one or more movements in the opposite direction of the desired muscle's movements. For example, if you want to stretch the left external oblique, you could perform a movement that involves trunk extension and right trunk lateral flexion. When a muscle has a high level of stiffness, you should use fewer simultaneous opposite movements (for example, to stretch a very tight external oblique, you could start by doing only trunk extension). As a muscle becomes loose, you can incorporate more simultaneous opposite movements.



Muscle	Trunk extension	Trunk flexion	Lateral trunk flexion
External oblique		✓	✓
Iliacus		✓	
Iliocostalis lumborum	✓		
Internal oblique		✓	✓
Interspinales	✓		
Intertransversarii			✓
Longissimus thoracis	✓		
Multifidus	✓		✓
Psoas major		✓	
Quadratus lumborum		✓	✓
Rectus abdominis		✓	
Rotatores	✓		✓
Spinalis thoracis	✓		